

VAR MODEL: Relationship between Real Estate Loans and Mortgage Rate

Xinyue Yan
Universtiy of Texas at San Antonio

Abstract

This project aims to develop a VAR model to explain the real estate loans by mortgage rate and other macroeconomic factors.

It is tried to make a suitable guess model by analyzing the data of real estate loans of all commercial bank 1980-2020, consumer price index of all items, short-term deposit rate as investment rate and lending rate which is mortgage rate of US. Frequency of time series dataset are monthly.

By taking difference of dataset and test Unit root to make sure the datasets are stationary. Durbin-Watson test for residual autocorrelation.

Though this project gives a guidance for policymakers to anticipate business cycles, as it conditions the right timing for their policies.

Introduction

This context is reviving research on credit. The recent literature, mainly empirical, generally seeks to find how the credit market has passed the financial shocks on to the real sector and develop macro-econometric models for explaining.

We first propose a parsimonious VAR model, which represents an intuitive way of modelling dynamic linkages between the key variables which contribute to explaining credit market conditions.

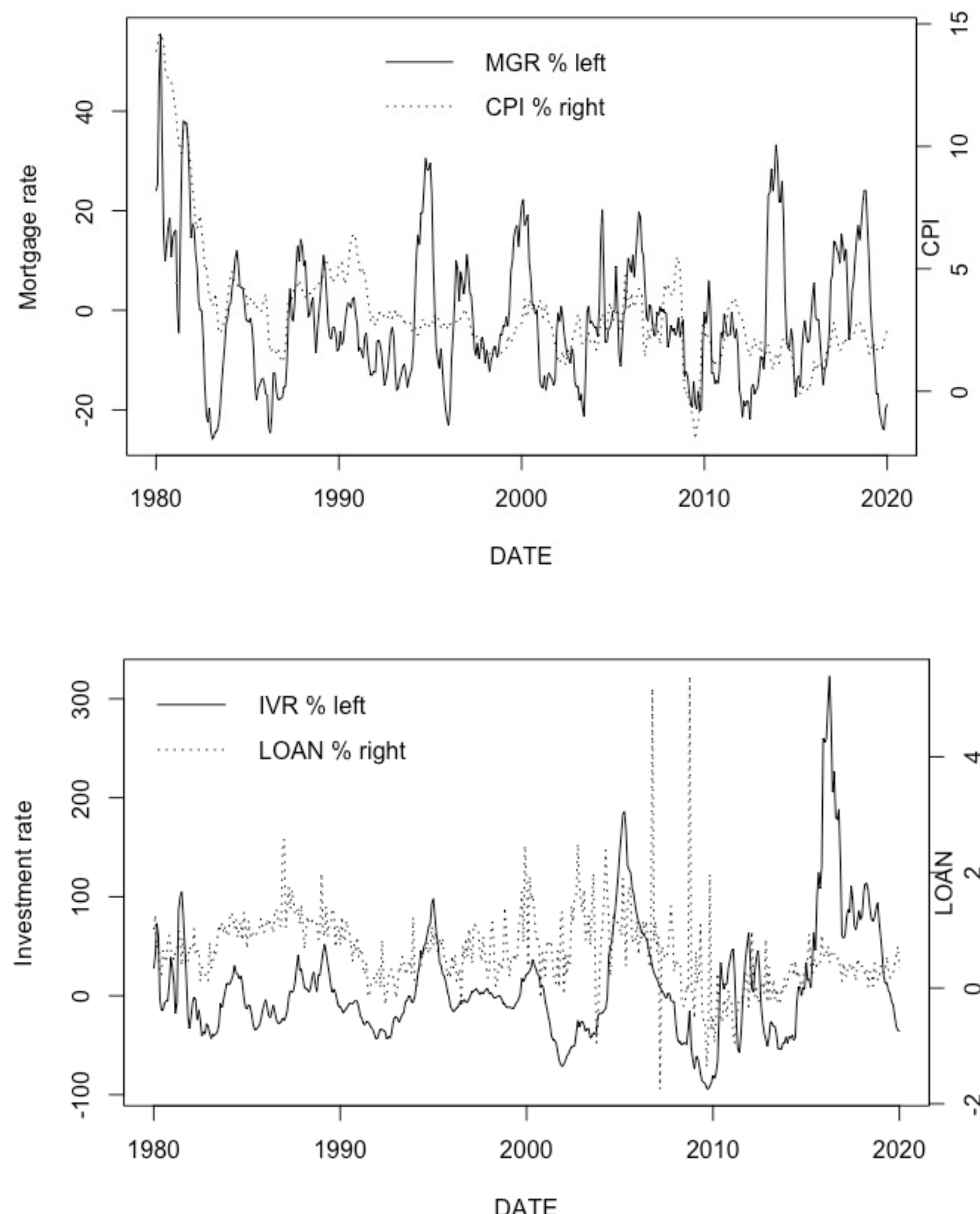
VAR model, defined as follows:

$$A_0 Z_t = \alpha + \sum_{i=1}^p A_i Z_{t-i} + \varepsilon_t$$

$$Z_t = (\Delta loan_t, cpi_t, ivr_t, mgr_t)$$

Where ε_t are structural shocks and is A_0 lower triangular $\begin{bmatrix} a & 0 & 0 & 0 \\ b & c & 0 & 0 \\ d & e & f & 0 \\ g & h & i & j \end{bmatrix}$.

Figure 1. Time series of Mortgage rate, Cpi, Investment rate and real estate loans(percentage chage)



Methods

Take difference and test unit root of 4 variables.

	$\Delta loan$	cpi	ivr	mgr
Dickey-Fuller	-3.1545	-5.3285	-4.4721	-7.6013
P VALUE:	0.02385	0.01	0.01	0.01

Test for VAR lag selection.

Schwarz criteria suggest $p = 2$.

Run VAR model with constant and lags equals to 2.

Test for residuals---Durbin-Waston Test.

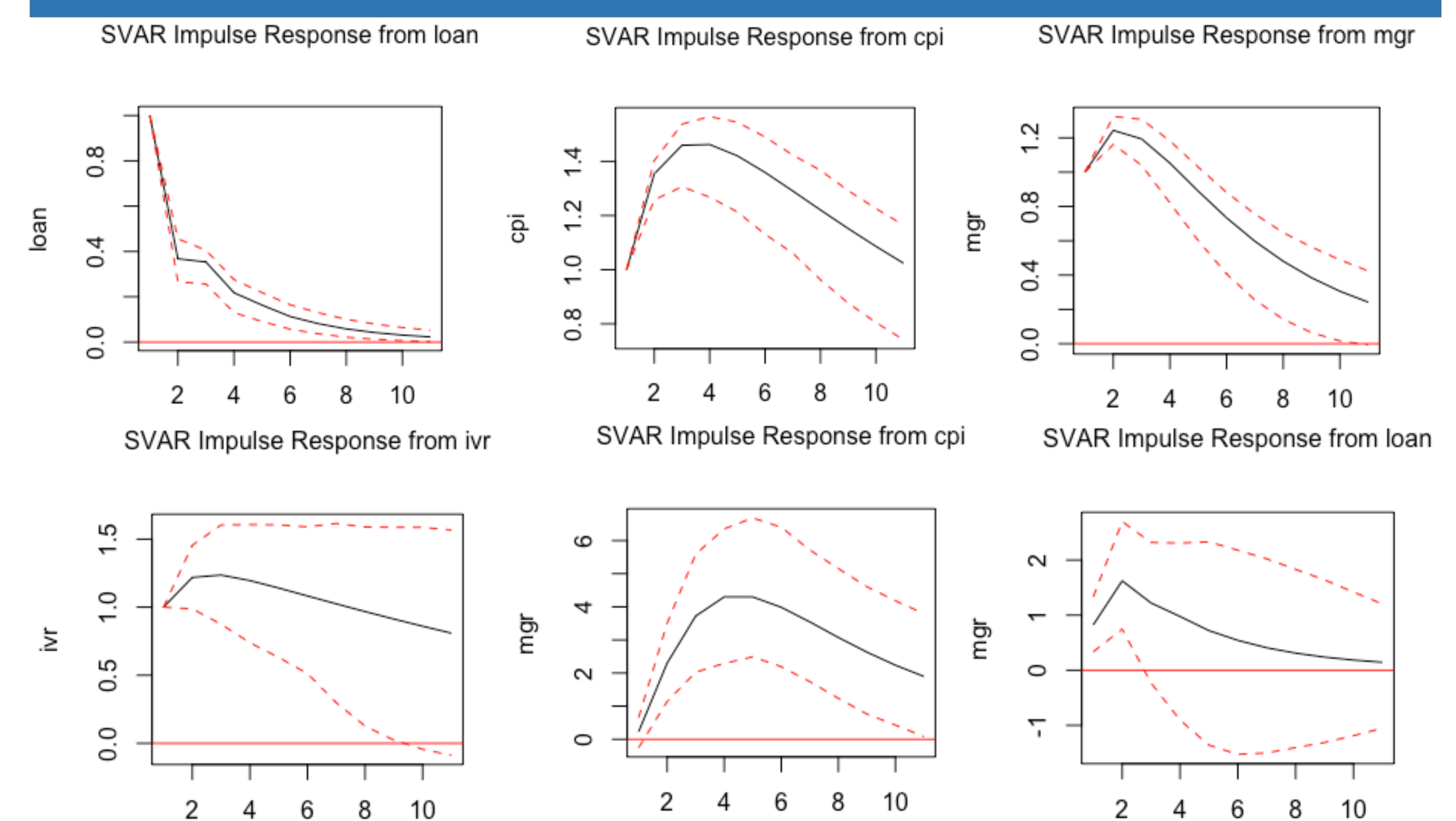
DW=1.91, residuals of VAR model are stationary white noises.

Make short-run restrictions.

Assume output is casually prior to inflation, investment rate and mortgage rate: if there is a shock to mortgage rate, it will take time to be reflected in rest of variables.

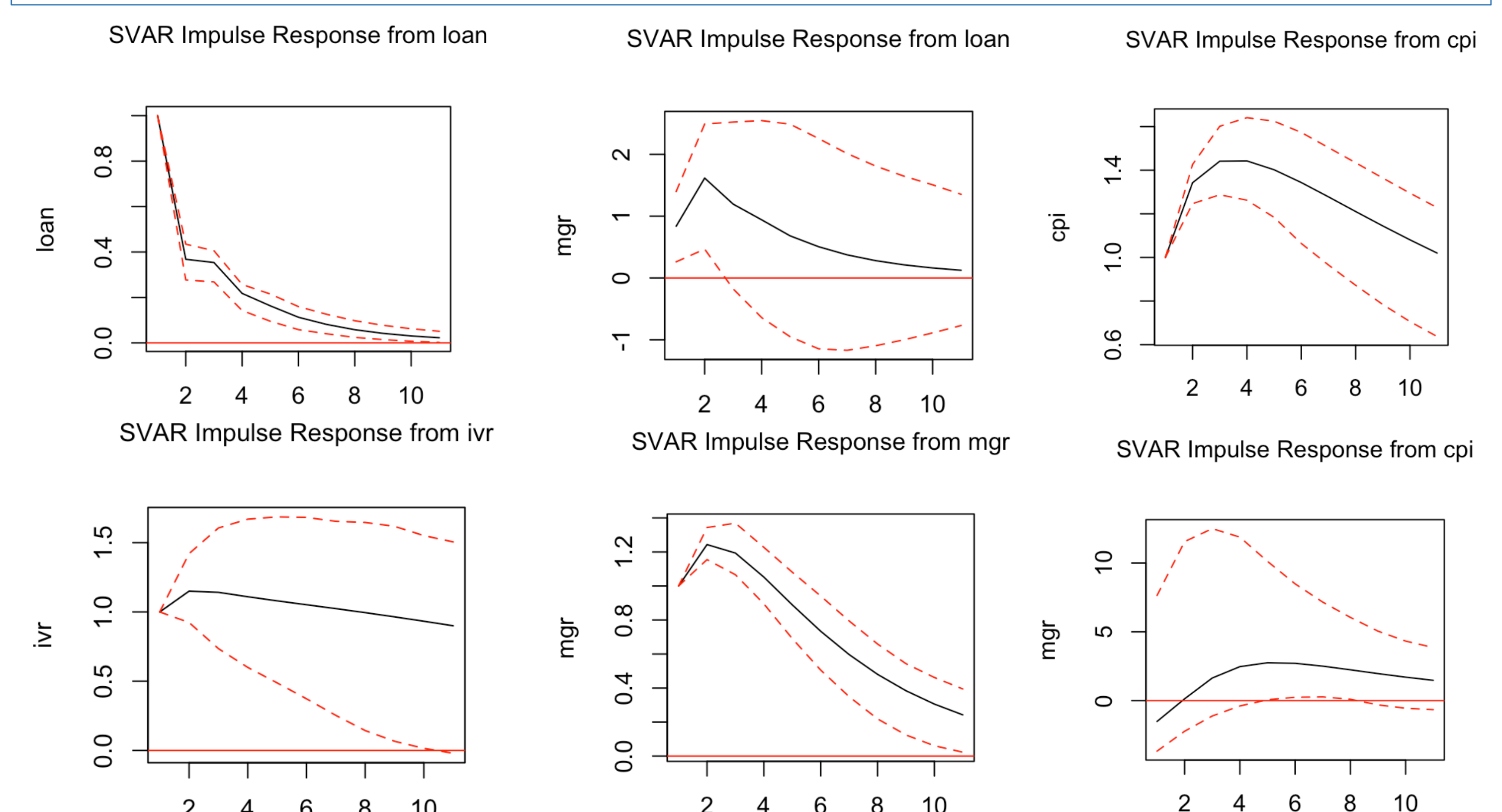
Run SVAR and plot IRFs.

Results



Robustness

Change the order of variables: $Z_t = (\Delta loan_t, ivr_t, cpi_t, mgr_t)$



Conclusions

- Loan has significant long effect with shock of itself.
- Inflation has strong significant long effect on shock of itself, but it has no significant effect on shock of loan.
- Mortgage rate has significant and presistent effect on shock of inflation and shock of itself. and it is contemporaneous effect (3 months) on shock of loan.
- Interest rate has significant effect on shock of itsel and it disappear 9 months later.
- Inflation dosen't have effect on shock of loan another variables, it is exdogeneous variables in this case.

Contact

<Xinyue Yan>
<UTSA win575>
Email: xinyue.yan@my.utsa.edu

References

Grégory Leveuge (2017) Explaining and forecasting bank loans. Good times and crisis, Applied Economics, 49:8, 823-843, DOI:10.1080/00036846.2016.1208350